

Michael Dowling

SENIOR DATA ENGINEER

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SUMMARY

Highly skilled **Senior Data Engineer** with over 9 years of experience designing and scaling cloud-native data platforms, real-time streaming pipelines, and high-performance **ETL** workflows across Fortune 500 companies. Proficient in **Python, Apache Spark, Airflow, Kafka**, and **AWS** (Redshift, S3, Glue, EMR), with a proven track record of reducing data latency, improving model accuracy, and enabling secure, compliant analytics at scale. Adept at cross-functional collaboration with data science, ML, DevOps, and product teams to drive data reliability, governance, and business impact through automation, monitoring, and infrastructure as code.

EXPERIENCE

Senior Data Engineer | *Jul 2023 - Present*
eBay | *San Jose, CA*

- Engineered real-time, scalable data pipelines using **Apache Spark, Airflow**, and **Kafka**, reducing data latency by 80% across critical e-commerce systems.
- Designed and implemented cloud-native data architecture on **AWS** (**S3, Redshift, Glue, Lambda, EMR**), powering petabyte-scale analytics for product performance and personalization.
- Optimized ETL workflows using **PySpark**, cutting processing time by 60% and improving data pipeline reliability across 15+ business domains.
- Integrated **monitoring and alerting systems** with **Prometheus, Grafana**, and **CloudWatch**, reducing incident detection time by 70%.
- Led the migration from batch to streaming data ingestion via **Kafka Streams**, enhancing real-time metrics tracking for buyer/seller behavior.
- Developed and maintained CI/CD pipelines using **GitHub Actions, Terraform**, and **Docker**, reducing deployment errors by 90%.
- Collaborated with **data science teams** to build high-quality training datasets, improving ML model performance by 35%.
- Architected schema registries and data contracts for shared datasets using **Avro** and **Protobuf**, improving data discoverability and governance.
- Enabled GDPR/CCPA compliance with automated **PII masking, IAM-based access controls**, and **VPC encryption**, maintaining 99.99% security audit score.
- Tuned **Airflow's LocalExecutor** and implemented DAG parallelization, achieving a 3x throughput increase during peak processing windows.
- Orchestrated nightly batch processes with failover and retry logic, ensuring 99.95% SLA adherence across all pipelines.
- Introduced robust data validation layers using **Great Expectations**, catching 90% of schema drift issues before production impact.
- Reduced AWS infrastructure costs by \$120K annually through resource optimization and auto-scaling policies across **Fargate** and **Kubernetes** clusters.
- Created Jupyter-based self-service analytics environment with secure access to sanitized production datasets.
- Conducted root cause analysis and triaged critical pipeline failures, reducing recovery time from 2 hours to under 10 minutes.
- Acted as technical lead for cross-functional initiatives; mentored junior engineers on Spark optimization, CI/CD, and modular ETL design.

Data Engineer | *Jan 2018 – Jun 2023*
Walmart | *Bentonville, AR*

- Designed and deployed large-scale **data pipelines** across merchandising and supply chain domains using **PySpark**, **Airflow**, and **AWS Glue**, supporting 10B+ daily events.
- Migrated on-prem Hadoop workloads to **AWS S3 + Athena**, cutting compute costs by 45% and improving data query speed by 4x.
- Implemented streaming ingestion with **Kafka** and **Flink**, enabling near real-time inventory updates and reducing stockout errors by 25%.
- Created modular ETL frameworks with schema validation and automated retries, reducing failure rates by 65%.
- Developed robust BI data models using **Redshift**, **SQL**, and **dbt**, enabling self-serve analytics and reducing ad hoc requests by 40%.
- Enhanced data integrity with automated data profiling, schema evolution handling, and proactive anomaly detection alerts.
- Partnered with analytics and ML teams to build training datasets for price optimization and customer churn prediction models.
- Led the implementation of **Delta Lake** architecture on S3, enabling ACID transactions, time travel, and scalable data versioning.
- Built out reusable **Terraform** modules for provisioning data infra (VPCs, IAM, Redshift clusters), reducing setup time by 80%.
- Managed continuous delivery pipelines with **Jenkins** and **GitHub Actions**, improving release frequency and quality.
- Conducted complex performance tuning on SQL queries and Spark jobs, reducing ETL runtime by over 50%.
- Enabled business users with dynamic **Power BI** dashboards backed by real-time data, driving informed decision-making across 10+ departments.
- Implemented centralized metadata and lineage tracking using **Amundsen** and **Glue Catalog**, increasing data trust and reuse.
- Achieved 100% compliance with internal audit requirements through version-controlled access policies and encrypted storage (S3 SSE-KMS).
- Automated onboarding of 30+ data sources using Python scripts, JSON schema validation, and REST API integrations.
- Created Airflow DAGs with SLA monitoring, dynamic task generation, and DAG parameterization for reusable orchestration patterns.
- Built custom plugins for **Airflow** and **Spark** to integrate with third-party APIs and legacy systems.

Data Analyst | *May 2015 – Dec 2017*
Kroger | *Cincinnati, OH*

- Designed ETL pipelines using **SSIS**, **Python**, and **SQL Server** to consolidate retail sales, inventory, and pricing data across 500+ stores.
- Built interactive dashboards in **Tableau** and **Power BI**, streamlining executive reporting and driving data-driven decisions.
- Automated weekly and monthly reporting workflows, saving 15+ hours/week and improving accuracy by 99.9%.
- Conducted advanced analytics using **SQL** and **Python** (Pandas, NumPy, Matplotlib) to uncover trends in consumer behavior and sales patterns.
- Developed customer segmentation models to support personalized marketing campaigns and loyalty program optimization.
- Partnered with supply chain and finance teams to model inventory turnover, shrinkage, and demand forecasting.
- Created KPIs to track store performance, pricing effectiveness, and promotional lift, resulting in 10% revenue increase in pilot stores.
- Migrated legacy reports to centralized dashboards with drill-down capabilities and real-time refresh.
- Conducted root cause analysis for revenue anomalies and inventory mismatches, improving reconciliation workflows.

- Integrated external data (weather, holidays, competitor pricing) to enrich decision-making models.
- Standardized metric definitions and reporting templates across departments, reducing reporting conflicts by 80%.
- Cleaned and validated 1TB+ datasets prior to data warehouse ingestion, improving data quality for BI consumption.

EDUCATION

Sep 2011 – Apr 2015

Bachelor's degree in Computer Science, Barry University

SKILLS

- **Languages/Querying:** Python, SQL, Scala, Bash
- **Big Data & Processing:** Apache Spark, Kafka, Flink, Hadoop, Delta Lake
- **Data Orchestration:** Apache Airflow, AWS Glue, Step Functions
- **Cloud & Infra:** AWS (S3, Redshift, EMR, Lambda, Fargate, EC2), Terraform, Docker, Kubernetes
- **Data Warehousing:** Redshift, BigQuery, Snowflake, MySQL, MongoDB
- **Data Quality & Testing:** Great Expectations, Pytest, dbt
- **Monitoring & CI/CD:** Prometheus, Grafana, CloudWatch, Jenkins, GitHub Actions
- **Visualization & BI:** Tableau, Power BI, Superset
- **Compliance:** PII handling, GDPR, CCPA, IAM Policies, Encryption at Rest (KMS, Vormetric)